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Research Paper

Do earnings events reset the trading clock?

Mike Lipkin, Arjun K. M. and Leon Tatevossian

Department of Finance and Risk Engineering, NYU Tandon School of Engineering,
6 MetroTech Center, Brooklyn, NY 11201, USA;
emails: mdl9606@nyu.edu, ak8196@nyu.edu, lt56@nyu.edu

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ABSTRACT

Larger-than-normal trading volumes of both stocks and vanilla options prior to, and just after, quarterly earnings announcements suggest there are profits to be mined from these singular events. Often a stock price will “trend into the announcement”. Does trending matter? For this exercise we examine a large collection of earnings events (approximately 14 000 observations, from three widely disparate discrete years) and extract the subset of outcomes for which the price strongly increased or declined into the earnings date. Our conclusion: no momentum is retained, either in the realized return over the announcement (which we label the “earnings return” or “jump”) or in the subsequent short-horizon price performance.

Keywords: price trending; earnings announcements; time scales; option implied volatility; price jumps.

1 INTRODUCTION

Momentum – more descriptively, trending stock prices – has been the subject of much theoretical and empirical analysis, with some recent work employing machine

Corresponding author: M. Lipkin

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learning (Avramov and Hore 2015; Gil *et al* 2024; Han 2025; Kaeley *et al* 2023; Lempérière *et al* 2014; Müller and Müller 2018; Zakamulin and Giner 2022; Zarattini *et al* 2025). This attention is not surprising: the assurance of "coarse grained" directionality (monotonicity) of prices in a densely traded temporal space could presumably guarantee a profit. We state "presumably" because it would be necessary to enter go-along trades and then exit while the trending continues. The recognition that, in the run-up to earnings, newly arrived agents get involved in the stock (with distinct preearnings strategies) means that the risk/reward of trend-linked decision-making can be examined game-theoretically. This research track was developed in recent work by Chriss (2024a,b,c, 2025).

The fact there is trending in the first place, possibly as a result of multiple cotemporal trading strategies, suggests the presence of an additional (hidden) timescale, which we denote by τ.[1] This timescale represents the "mean trending duration". Equivalently, τ is the interval after which, on an ex ante basis, longer-lived positions would convergently return no better than a codirectional aggregate market (eg, the SPDR Standard & Poor's 500 Exchange-Traded Fund Trust) or a more tailored tracking position. This scale is not directly observable because it is arguably a dynamical consequence of multiple coexisting trading strategies.

A natural question is whether a trending regime includes events not described by the regular noise process. This means that, from the aggregate perspective, a bias could materialize in the "earnings returns" for the "trending subset" of observations. In an earlier paper we showed that a symmetric, fat-tailed set of outcomes characterize the earnings returns (Arjun K. M. *et al* 2025). If the same were true for trending stocks, then no earnings-event-only arbitrage would be achievable in a directed position. The earnings event would reset the clock, so to speak.

If, however, trending "passes through" the earnings event, it becomes a serious matter to understand why we accept the martingale hypothesis, and to understand why trading does not extinguish what seems to be a perpetual-motion money-making machine. Note that these two scenarios are independent questions. It is possible to imagine symmetric, fat-tailed jumps with the earnings announcement followed by a resumption of trending or antitrending. If trending passed through the earnings announcement, an alert trader could simply enter a market order to buy or sell the postearnings opening (as appropriate).

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[1] Derivatives pricing depends on the stochastic process for the price evolution. In the context of this operative timescale it is possible to image a uniform background process reflecting as many factors as desired (priced by a favorite model: geometric Brownian motion, Heston, stochastic alpha–beta–rho, etc) plus an additional directed noise, characterized by τ.

2 APPROACH

Following the framework of our earlier paper (Arjun K. M. *et al* 2025), we consider three discrete, disparate years (2009, 2014 and 2019) and more than 4000 earnings events per annum. From these observations we select those occurrences with at least a $\pm 10\%$ move over the three-week calendar period ahead of the earnings date. The choice of a 10% threshold is somewhat arbitrary;[2] it is chosen to ensure a sizable trend, as well as to guarantee that not too large a set of occurrences remains.

The selection of the three-week interval corresponds empirically to the appearance of higher short-expiry at-the-money implied volatility, distinguishable from background long-term (typically 180-day) implied volatility.[3] For this data subset we examine the distribution of jumps and the subsequent five-day price move following the announcement.

We discern a complete dissociation of the earnings return from the prior trending. Just as with the entire collection of events, this subset of "strongly trending into earnings" stocks experiences symmetric, fat-tailed jumps, with no evidence of memory. We also examine the five-day postearnings performance. Along with the fact that earnings returns are agnostic to prior trending, we find that the price action in the subsequent period retains no memory. The conclusion is straightforward: earnings reset all clocks. The hidden timescale (τ) does not survive the earnings release.

Section 3 describes our methodology. The results are provided in Section 4. Section 5 provides a short working example and summarizes our findings.

3 METHODOLOGY

Earnings announcement dates/times for the three calendar years were acquired from a reliable source for stocks listed on US exchanges (Nasdaq and the New York Stock Exchange).[4] The total number of stocks per year exceeded 1000. We used Bloomberg to acquire pricing data for the four quarterly earnings announcements, following each of the stocks from three weeks before the earnings date to five days after.

To concentrate entirely on trending stocks, we calculated the price return over the prior three weeks and included only the cases exceeding a 10% change in magnitude. As noted, the choice of 10% was somewhat arbitrary, made to ensure a size that is reasonable for statistical purposes but far from the entire set (which we have already shown exhibits no bias (Arjun K. M. *et al* 2025)).

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[2] Too large a universe would converge to the symmetric, fat-tailed distribution we saw in Arjun K. M. *et al* (2025). Too small a sample size would be statistically indistinguishable from noise.

[3] Based on unpublished work by Mike Lipkin.

[4] Wall Street Horizon, a TMX company (www.wallstreethorizon.com).

To investigate the second question, we followed this subset of cases from the postearnings opening to the close five trading days later.[5] We note also that, while an approaching earnings event reveals itself in the form of increasing short-expiry implied volatility up to three weeks in advance, option-implied volatilities rapidly revert to baseline levels, often within one trading day.[6]

We fit the earnings-return histogram to the Laplace distribution, consistent with our prior work (Arjun K. M. *et al* 2025). We fit the five-day postearnings returns to the Student t distribution.

4 RESULTS

When conceptualizing trading strategies it is natural to separate the trending-up and trending-down examples. Out of the three-year total of 13 946 earnings events, the trending-stocks constraint resulted in 1828 upward-trend cases (13.1% of the total) and 952 downward-trend cases (6.8%).[7]

Table 1 and Figures 1–4 show the empirical distribution of the earnings returns (separated into down and up preearnings trenders) and the approximate fit to the Laplace distribution. In each case (down and up) the histogram exhibits a positive mean; the means are roughly equal, with both less than 0.50%. Given the modest magnitude of the means in comparison with the corresponding standard deviations (7.32% and 6.35%, respectively), we work from the assumption that the nonzero means are artifacts of the data set.

For this same set of observations Table 2 and Figures 5–8 display the five-day postearnings returns, again divided into down and up trenders. For these return realizations we examine the fit to the Student t distribution.

The histograms of the five-day postearnings returns share some of the statistical attributes of the earnings-return data. Both mean values are modest in relation to the standard deviations, with a negative mean (–0.53%) for the down trenders and a positive mean (1.89%) for the up trenders. One notable difference is that, when normalized using the corresponding standard deviation, the mean is much closer to zero for the preearnings down trenders.

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[5] The choice of five trading days is designed to be short enough that noise does not wash out any trending signals and long enough to capture the reappearance of any trending; it also provides some assurance that we do not include subsequent one-off events that have outsized impacts on the stock price.

[6] Based on unpublished work by Mike Lipkin.

[7] The result for individual years was similarly unremarkable but noisier.

TABLE 1 Earnings returns.

	Trending down pre earnings	Trending up pre earnings
Observations	952	1828
Mean (%)	0.36	0.42
SD (%)	7.32	6.35
10th percentile point (%)	−6.8	−6.4
90th percentile point (%)	9.2	7.5
2 SD value (left tail) (%)	−14.3	−12.3
2 SD value (right tail) (%)	15.0	13.1
Observations outside 2 SD	5.15% of total (952)	5.63% of total (1828)
Fitted Laplace parameters		
Location (mean) (%)	0.23	0.27
Scale (%)	4.92	4.44
SD (%)	6.96	6.28
Kolmogorov–Smirnov statistic	0.0343	0.0161

SD, standard deviation. The Kolmogorov–Smirnov statistic, usually denoted D (or D_n, where n is the sample size), is the maximum vertical distance between the data's (empirical) cumulative density function (cdf) and the cdf of the target parametric distribution. The maximum is taken for all values in the support of the empirical cdf. The statistic is evaluated with respect to a confidence level on the alignment of the distributions, with the characterization dependent on the number of observations.

FIGURE 1 Earnings returns (negative trend).

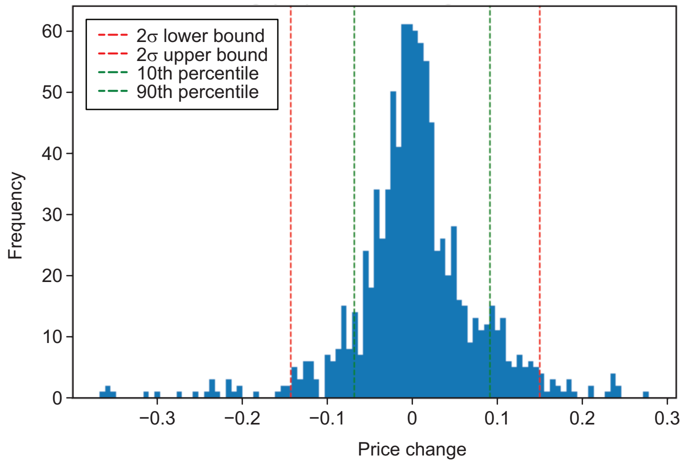
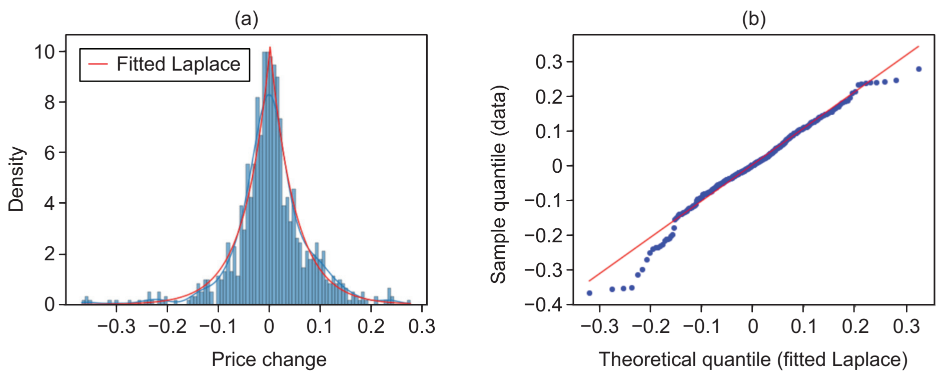
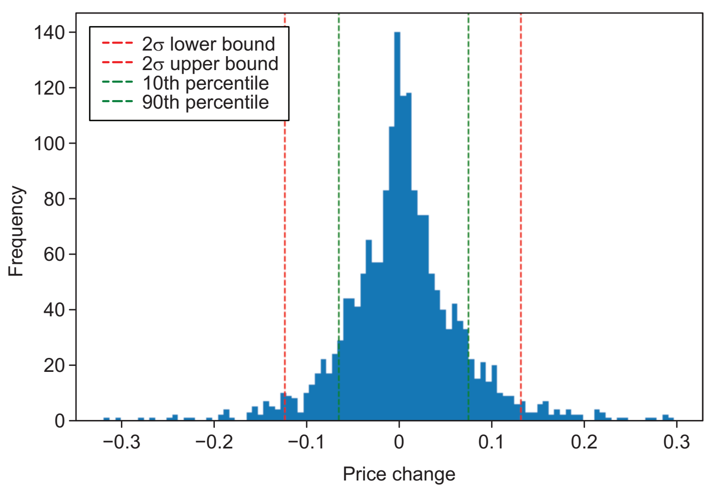


FIGURE 2 Earnings returns fitted to Laplace (negative trend).



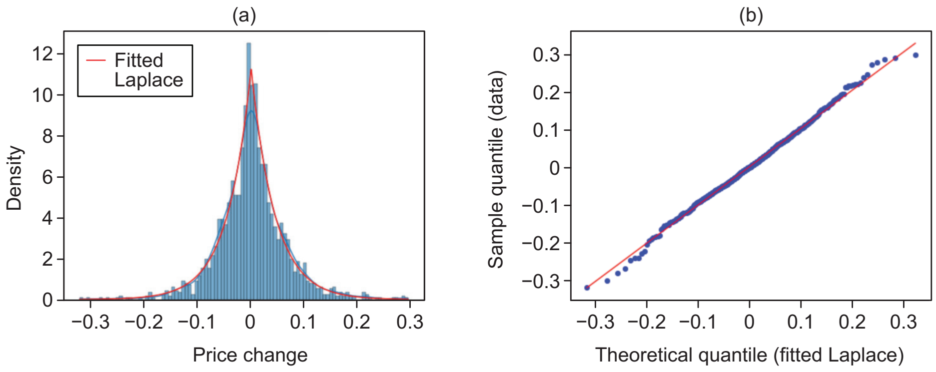
(a) Fitted Laplace distribution versus histogram. (b) Quantile–quantile plot.

FIGURE 3 Earnings returns (positive trend).



An additional remark might clarify a potential misunderstanding. In our earlier paper we found that the Laplace distribution was the best (though imperfect) fit to model the earnings-return histogram (Arjun K. M. *et al* 2025). Unsurprisingly, that distribution fits this restricted set of earnings returns (selected from the strong trenders). Our prior paper was purely empirical; we did not offer a theoretical model

FIGURE 4 Earnings returns fitted to Laplace (positive trend).



(a) Fitted Laplace distribution versus histogram. (b) Quantile–quantile plot.

producing such a distribution. In our view, generating it from first principles is an open problem.

On the other hand, the five-day postearnings returns for the stocks would not, a priori, be expected to follow the same behavior, given that this interval reflects an updated set of price and risk determinants. In fact, we found a superior fit using the Student t distribution (consistent with the work of Plerou *et al* (1999) and Ma and Serota (2014)).[8]

5 SUMMARY

It is reasonable to believe that the universe of traders/investors consists of well-capitalized firms, individual small traders and everything in between. It is also plausible that some traders are very knowledgeable while others may be punters. Option and stock volume in the run-up to earnings strongly suggests that traders are motivated to place or remove bets within this window.

To provide some color, here we take as an arbitrary example the case of Apple Inc. around its earnings date of January 30, 2025 after market close. In the five days

[8] More precisely, we are fitting the empirical distribution to the location–scale t distribution. This distribution's parameters are usually denoted by (μ, τ^2, ν) for the location, the square of the scale (τ) and the degrees of freedom, respectively. The associated random variable R satisfies $R = \mu + \tau S$, where S is a Student t variable with ν degrees of freedom, and thus $\mathrm{Var}[R] = \tau^2\nu/(\nu - 2)$.

TABLE 2 Five-day postearnings returns.

	Trending down pre earnings	Trending up pre earnings
Observations	952	1828
Mean (%)	−0.53	1.89
SD (%)	13.23	9.86
10th percentile point (%)	−16.1	−8.7
90th percentile point (%)	14.9	12.8
2 SD value (left tail) (%)	−27	−17.8
2 SD value (right tail) (%)	25.9	21.6
Observations outside 2 SD	6.20% of total (952)	5.85% of total (1828)
Fitted Student t parameters		
Degrees of freedom	4.13	3.39
Location (mean) (%)	−0.42	1.42
Scale (%)	9.89	6.68
SD	13.77	10.43
Kolmogorov–Smirnov statistic	0.0225	0.0183

SD, standard deviation.

FIGURE 5 Five-day postearnings returns (negative trend).

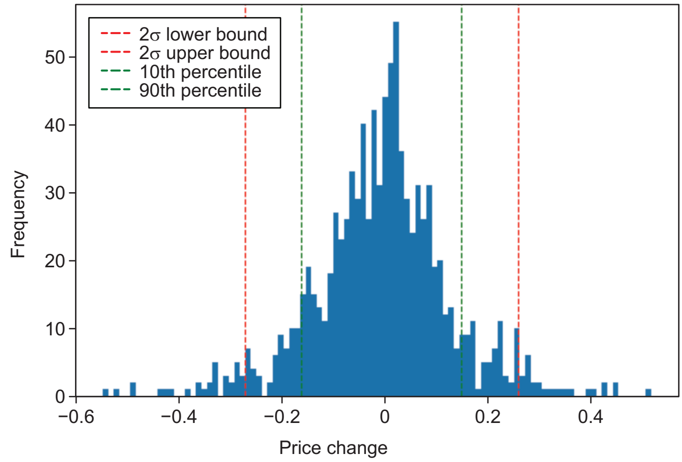
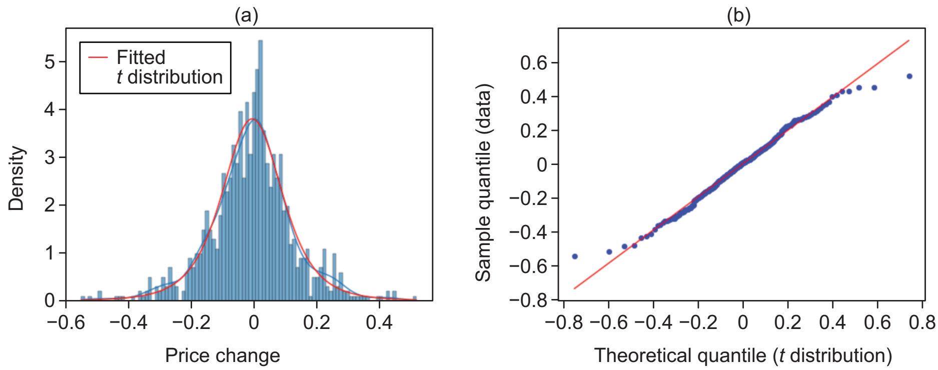
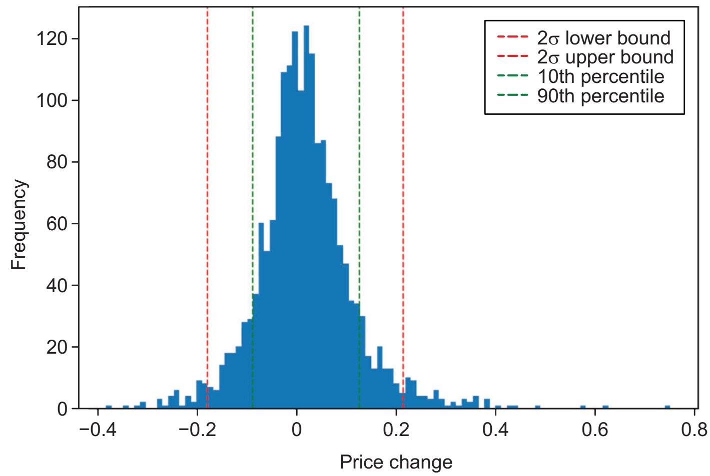


FIGURE 6 Postearnings returns fitted to Student t (negative trend).



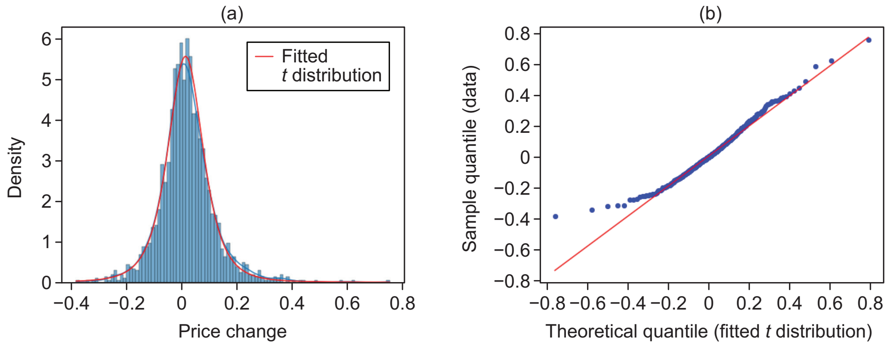
(a) Fitted t distribution versus histogram. (b) Quantile–quantile plot.

FIGURE 7 Five-day postearnings returns (positive trend).



prior to earnings (January 26–30), daily stock and option volume averaged 65 million shares and 930 000 contracts, respectively. The January 31 volume was 101 million shares and 2 630 000 contracts. In contrast, over the subsequent four days volume averaged 47 million shares and 810 000 contracts (substantially smaller). This is typical.

FIGURE 8 Postearnings returns fitted to Student t (positive trend).



(a) t distribution versus histogram. (b) Quantile–quantile.

The empirical work described in this note appears to suggest that, absent any idiosyncratic attributes for the stock's price determinants, a naive directional earnings-event-based strategy cannot return a profit. Within the subset of trending stocks, no directionality is observed in the earnings return. Further, any trend is extinguished by the earnings event.[9] This study suggests that profit expectations while trading just prior to earnings are subject to colorless randomization as positions are carried through the event.

DECLARATION OF INTEREST

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

ACKNOWLEDGEMENTS

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[9] In our prior paper (Arjun K. M. *et al* 2025) we conducted the corresponding study for our entire three-year universe of earnings events.

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